QT-02 Quantum Computing Theory & Practices

L	T	P	C
3	0	2	4

Prerequisite knowledge: fundamental physics, mathematics and programming skills like Python and experience with simulation tools.

Course description and objectives: This course covers the basics of quantum technology and quantum computation. It also emphases quantum logic gates, circuit models, and key quantum algorithms.

Module-1

24L+0T+16P = 40 hours

Unit-1

Linear algebra for quantum mechanics; Heisenberg's matrix representation; Quantum states and Hilbert space; The Bloch sphere; no-cloning theorem; Density Matrices and Mixed States.

Unit-2

Complex vector spaces; bra and ket notation; Physical observables and Hermitian operators; Unitary operators; expectation values; Unitary Transformations; Operators for density, momentum and kinetic energy; time dependent Schrodinger wave equation; Trapped atoms and ions; superconductivity, Meissner effect, Cooper pairs and energy gap-BCS Theory; DC & AC Josephson Junction; superconducting qubits.

EXPECTED PRACTICES

- ❖ Symbolic/numeric validation of operator properties (Hermitian and Unitary Operators Verification)
- ❖ Simulate mixed vs pure states and density matrix evolution (Density Operator Visualization)
- ❖ Simulate Bell states and measure correlation (Entangled States and Correlations)
- ❖ Use IBM Qiskit or Quantum Inspire to create circuits (Basic Quantum Circuit Builder)

Module-2

24L+0T+16P = 40 hours

Unit-1

Artificial atoms using superconducting circuits; Semi-conducting Quantum dots; Spin half systems and photon polarizations; Solovay - Kitaev Theorem; Quantum advantage claims; Quantum correlations and Entanglement; Review of Turing machines; classical computational complexity-Time and space complexity (P, NP, PSPACE).

Unit-2

Quantum operations; Hadamard, Pauli X, Y, Z gates, CNOT, phase gates, Pauli X, Y, Z; measurements; Unitary gate operations as time evolution of multi-qubit systems; Bell theorems &

measurement; Overview of Quantum Circuits; Deutsch algorithm; Deutsch Josza algorithm; Database search- Grover's algorithm; Shor's Algorithm.

EXPECTED PRACTICES

- Build H-Gate and observe interference (Superposition and Interference in Qubits)
- Simulate Bell inequality violation in entangled systems (Bell Inequality Violation Test)
- Simulate or demonstrate polarization states as qubits (Photon Polarization States)
- Simulate energy levels in Josephson junction systems (Artificial Atoms in Superconducting Circuits)
- Test CHSH inequality using quantum circuits (Bell's Inequality Violation)
- Time complexity demo using simple sorting/searching (Classical Algorithm Complexity (P vs NP))
- Simulate and test single-qubit (H, X, Y, Z) and two-qubit gates (CNOT, CZ) (Quantum Gate Implementation)
- Create custom circuits to test gate ordering and qubit entanglement (Quantum Circuit Design)
- Use {H, T, CNOT} to build arbitrary quantum gates (Universal Gate Set Verification)
- Implement and verify single-bit problem resolution with 1 query (Deutsch Algorithm)
- Differentiate between constant and balanced functions (Deutsch-Jozsa Algorithm)
- Demonstrate quantum speedup in database search (Grover's Algorithm)

CO No.	Course Outcomes	Blooms level	Module No.	Mapping with POs
1	Apply linear algebra tools, Heisenberg's formalism, and quantum state representations like the Bloch sphere.	Apply	3	1(3), 2(2)
2	Analyze operators in quantum mechanics, their physical significance, and time-dependent evolution via Schrödinger's equation, and key concepts of superconductivity including Cooper pairs, BCS theory, and Josephson junctions.	Apply, Analyze	3	1(3), 2(3)
3	Describe artificial atoms using superconducting circuits and understand the Solovay-Kitaev theorem. Also Compare classical and quantum computational models including Turing machines, complexity classes, and entanglement.	Analyze	4	1(3), 2(3), 4(3)

4	Design and implement quantum logic circuits using quantum gates and apply them to known algorithms. Analyze and simulate algorithms such as Deutsch-Jozsa, Grover's, and Shor's to demonstrate quantum speed-up.	Analyze	4	1(3), 2(3), 5(2), 11(2)

Text books:

1. Introduction to Quantum Mechanics, David J. Griffiths, Cambridge University Press, 3rd edition, 1018.
2. An introduction to quantum computing P. Kaye, R. Laflamme and M. Mosca, (Oxford Univ Press, 2010)

Course References:

1. Quantum Mechanics for Engineers and Material scientists, P. Anantram and Daryoush Shiri, World scientific publishing company, 2023.
2. Chris Bernhardt, Quantum Computing for Everyone, The MIT Press, Cambridge, 2020.
3. David McMahon-Quantum Computing Explained-Wiley-Interscience, IEEE Computer Society (2008)
4. Quantum Information Science – Manenti R., Motta M., 1st Edition, Oxford University Press (2023)
5. Quantum computation and quantum information – Nielsen M. A., and Chuang I. L., 10th Anniversary edition, Cambridge University Press (2010)
6. A Pathak, Elements of Quantum Computation and Quantum Communication, Boca Raton, CRC Press (2015)
7. Quantum error correction and Fault tolerant computing, Frank Gaitan, 1st edition, CRC Press (2008)
8. Quantum computing explained, David McMahon, Wiley (2008)